

FAKHAR STEM SPHERE

Comprehensive Chapter Test

Class 9 Mathematics (FBISE)

Chapter 4: Factorization and Algebraic Manipulation

Name:	Roll No.:	Date:

Suggested Time: 3 Hours

Total Marks: 95

Chapter Test

General Instructions: Attempt all questions. Show complete working in Sections B and C. Calculations and algebraic steps should be written clearly. This paper is designed for FBISE-style chapter preparation.

SECTION A - OBJECTIVE

15 Marks

Q.1. Choose the correct option. Each MCQ carries 1 mark.

1. For two polynomials A and B , $HCF \times LCM =$

- | | |
|---------------|------------------|
| (A) $A + B$ | (B) $A - B$ |
| (C) $(HCF)^2$ | (D) $A \times B$ |

2. Which is the highest common factor of $-12x^2y^2$, $6xy^3$, $24x^2y^2$?

- | | |
|-------------|-----------------|
| (A) $-6xy$ | (B) $6x^2y^3$ |
| (C) $6xy^2$ | (D) $-24x^2y^3$ |

3. What is the least common multiple of $7x - 6xy$ and $5xy^3 - 3x^2$?

- | | |
|---------------------------|--------------------------------|
| (A) $(7 - 6y)(5x^3 - 3x)$ | (B) $(7x - 6xy)(5y^3x - 3x^2)$ |
| (C) $x(7 - 6y)$ | (D) $x(7 - 6y)(5y^3 - 3x)$ |

4. Factorization of $4a^2 - 12ab + 9b^2$ is:

- | | |
|-------------------|-------------------|
| (A) $(2a - 3b)^2$ | (B) $(2a + 3b)^2$ |
| (C) $(4a + 3b)^2$ | (D) $(4a - 3b)^2$ |

5. HCF of $3x^2$ and $6x^3$ is:

- | | |
|------------|------------|
| (A) $3x$ | (B) $6x^2$ |
| (C) $3x^2$ | (D) $3x^3$ |

6. Factorization of $49a^2 - 28ab + 4b^2$ is:

- | | |
|-------------------|--------------------|
| (A) $(7a + 2b)^2$ | (B) $(49a - 4b)^2$ |
| (C) $(7a - 2b)^2$ | (D) $(7a - 28b)^2$ |

SECTION A - OBJECTIVE (Continued)

15 Marks

7. What is the square root of $\left[- \left(2x + \frac{1}{x} + 1 \right) \right]^2$?
- (A) $\pm \left(2x + \frac{1}{x} + 1 \right)^2$ (B) $\left(2x + \frac{1}{x} + 1 \right)^2$
 (C) $2x + \frac{1}{x} + 1$ (D) $\sqrt{2x + \frac{1}{x} + 1}$
8. If the product of HCF and LCM of two polynomials is $(x^3 - y^3)(x + y)$, then what will be the product of these polynomials?
- (A) $x^4 - y^4$ (B) $(x - y)(x^2 + y^2)$
 (C) $(x^2 - y^2)(x^2 + xy + y^2)$ (D) $(x^2 - y^2)(x^2 - xy + y^2)$
9. What is the square root of $(15x^2 - 7y^2)^4$?
- (A) $\pm(15x^2 - 7y^2)^2$ (B) $\pm(15x^2 - 7y^2)$
 (C) $\pm(15x - 7y)^2$ (D) $\pm(15x - 7y)$
10. What is the difference of $\frac{-3x}{x + y}$ and $\frac{x}{x + y}$?
- (A) $\frac{-2x}{x + y}$ (B) $\frac{-3x - x}{2x + 2y}$
 (C) $\frac{3x^2}{x + y}$ (D) $\frac{-4x}{x + y}$
11. Factors of $-2 - a + a^2$ are:
- (A) $(a - 2)(a - 1)$ (B) $(a + 1)(a + 2)$
 (C) $(a + 2)(a - 1)$ (D) $(a + 1)(a - 2)$
12. Factorization of $x^2 - x + \frac{1}{4}$ is:
- (A) $\left(x + \frac{1}{2} \right) \left(x - \frac{1}{2} \right)$ (B) $\left(x + \frac{1}{2} \right) (x - 1)$
 (C) $\left(x - \frac{1}{2} \right) (x + 1)$ (D) $\left(x - \frac{1}{2} \right) \left(x - \frac{1}{2} \right)$
13. If the product of two polynomials is $(a - b)^2(a^2 + ab + b^2)$ and their HCF is $a - b$, what is their LCM?
- (A) $a^3 - b^3$ (B) $(a - b)^3(a^2 + ab + b^2)$
 (C) $(a - b)^2(a^2 + ab + b^2)$ (D) $a^2 + ab + b^2$
14. What is the highest common factor of $7x - 6xy$ and $5xy^3 - 3x^2$?
- (A) $(7 - 6y)(5y^3 - 3x)$ (B) $(7x - 6xy)(5y^3x - 3x^2)$
 (C) x (D) $x(7 - 6y)(5y^3 - 3x)$
15. Factors of $3x^2 - x - 2$ are:
- (A) $(x + 1), (3x - 2)$ (B) $(x + 1), (3x + 2)$
 (C) $(x - 1), (3x - 2)$ (D) $(x - 1), (3x + 2)$

SECTION B - SHORT QUESTIONS**40 Marks****Q.2. Attempt all parts. Each part carries 4 marks.****(i) Simplify:**

$$\frac{5}{5+p-18p^2} - \frac{2}{2+5p+2p^2}$$

[4 Marks]**(ii) Find the HCF of the polynomials**

$$x^3 + 2x^2 - 4x - 8 \quad \text{and} \quad 2x^3 + 7x^2 + 4x - 4.$$

[4 Marks]**(iii) Factorize the quadratic expression:**

$$3x^2 + 16x - 12.$$

[4 Marks]**(iv) For the algebraic expressions $P = 6x^2y$ and $Q = 9xy^2$, show that**

$$P \times Q = \text{LCM} \times \text{HCF}.$$

[4 Marks]**(v) Factorize:**

$$(x^2 - 7x + 4)(x^2 - 7x + 5) - 2.$$

[4 Marks]**(vi) If $A = 4a^2b$ and $B = 6ab^2$, verify that**

$$A \times B = \text{LCM} \times \text{HCF}.$$

[4 Marks]

(vii) Factorize:

$$x^3 + x^2 - 4x - 4.$$

[4 Marks]

(viii) Find the HCF and LCM of

$$x^2 - 9 \quad \text{and} \quad x^2 - x - 6.$$

[4 Marks]

(ix) Find the square root of

$$4x^4 + 12x^3 + 29x^2 + 30x + 25.$$

[4 Marks]

(x) Factorize:

$$(x - 2)(x - 6)(x - 3)(x - 4) - 2x^2.$$

[4 Marks]

SECTION C - LONG QUESTIONS

40 Marks

Q.3. Attempt all parts. Each part carries 8 marks. Show complete working.

(i) To convert

$$x^4 + 4x^3 + 10x^2 + 10x + 5$$

into a perfect square:

- What should be added to it?
- What should be subtracted from it?

[8 Marks]

(ii) Find the values of p and q if

$$x^4 - 12x^3 + px + q$$

is a complete square.

[8 Marks]

SECTION C - LONG QUESTIONS (Continued)**40 Marks**

(iii) For what value of k does the expression

$$y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$$

become a perfect square?

[8 Marks]

(iv) Find the HCF and LCM in simplified form using the factorization method:

$$2x^3 - x^2 - 8x + 4 \quad \text{and} \quad 2x^3 + 3x^2 - 8x + 3.$$

[8 Marks]

SECTION C - LONG QUESTIONS (Continued)**40 Marks**

(v) To make

$$a^4 - 10a^3 + 27a^2 - 9a + 2$$

a perfect square:

- i. What should be added to it?
- ii. What should be subtracted from it?
- iii. What will be the value of a ?

[8 Marks]**— End of Paper —**

Best wishes for your FBISE examination preparation.